JavaScript

BIS605

Software Development Technologies for Digital Business

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Resources

- https://www.w3schools.com
- https://www.geeksforgeeks.org
- https://www.w3docs.com/
- https://www.tutorialspoint.com/html

JavaScript

- **JavaScript** is a lightweight **programming** language.
- It is very easy to implement because it is integrated with HTML.
- It is open and cross-platform.
- JS focuses on user interfaces and interacting with a document.
- JS code runs on the client's browser

Syntax

- JavaScript (JS) can be implemented using JavaScript statements that are placed within the **<script>... </script>** HTML tags in a web page.

- The <script> tag can be anywhere in the web page but it is recommended to be within the <head> tag.

Syntax

- The script tag takes two important attributes –
  - **Language** – This attribute specifies what scripting language you are using.
  - **Type** – This attribute is what is now recommended to indicate the scripting language in use and its value should be set to "text/javascript".

```
<script language = "javascript" type = "text/javascript">
   JavaScript code
</script>
```

- We can link to JS files as well by using attribute src

```
<script src="myscripts.js"></script>
```

Syntax

- Whitespace and Line Breaks are ignored. You can use spaces, tabs, and newlines freely in your program.
- Semicolons are optional

```
<script language = "javascript" type = "text/javascript">
   <!--
      var1 = 10
      var2 = 20
   //-->
</script>
```

```
<script language = "javascript" type = "text/javascript">
   <!--
      var1 = 10; var2 = 20;
   //-->
</script>
```

- Case sensitive

Datatype and Variable

- Three primitive data types
  - Numbers
  - String
  - Boolean
- Variable in JS can be created to hold a value of any datatype. Variables are declared with the **var** keyword.

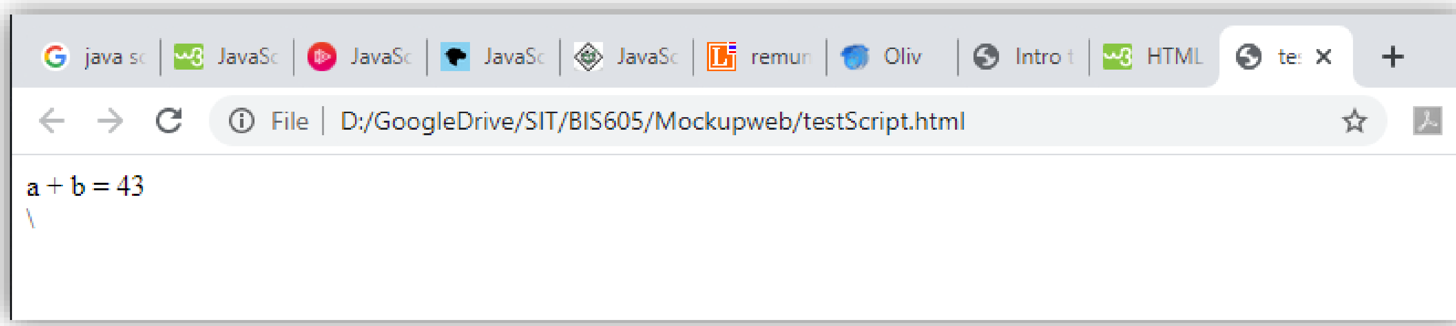
```
<script type = "text/javascript">
   <!--
      var money;
      var name;
   //-->
</script>
```

```
<script type = "text/javascript">
   <!--
      var name = "Ali";
      var money;
      money = 2000.50;
   //-->
</script>
```

Operators

- Arithmetic operators
  - + - * / %
- Comparison operators
  - == != > < >= <=
- Logical operators
  - && || !

```html
<!DOCTYPE html>
<html>
<body>
    <script type = "text/javascript">
        <!--
           var a = 33;
           var b = 10;
           var c = "Test";
           var linebreak = "<br />";

           document.write("a + b = ");
           result = a + b;
           document.write(result);
           document.write(linebreak);
        </script>
</body>\
</html>
```



If-Else

- JS supports the following forms of if…else statement
  - if statement
  - if…else statement
  - if….else if…. statement

```
var book = "maths";
if( book == "history" ) {
   document.write("<b>History Book</b>");
} else if( book == "maths" ) {
   document.write("<b>Maths Book</b>");
} else if( book == "economics" ) {
   document.write("<b>Economics Book</b>");
} else {
   document.write("<b>Unknown Book</b>");
}
```

```
var age = 20;

if( age > 18 ) {
   document.write("<b>Qualifies for driving</b>");
}
```

```
var age = 15;

if( age > 18 ) {
   document.write("<b>Qualifies for driving</b>");
} else {
   document.write("<b>Does not qualify for driving</b>");
}
```

While Loop

- The most basic loop in JavaScript is the **while** loop.
- The purpose of a **while** loop is to execute a statement or code block repeatedly as long as an **expression** is true.

```
var count = 0;
document.write("Starting Loop ");

while (count < 10) {
   document.write("Current Count : " + count + "<br />");
   count++;
}

document.write("Loop stopped!");
```

For Loop

- The '**for**' loop is the most compact form of looping. It includes the following three important parts:
  - The **loop initialization** where we initialize our counter to a starting value.
  - The **test statement** which will test if a given condition is true or not.
  - The **iteration statement** where you can increase or decrease your counter.

```
var count;
document.write("Starting Loop" + "<br />");

for(count = 0; count < 10; count++) {
   document.write("Current Count : " + count );
   document.write("<br />");
}
document.write("Loop stopped!");
```

Function

- A function is a group of reusable code which can be called anywhere in your program.

```
<head>
   <script type = "text/javascript">
      function sayHello() {
         document.write ("Hello there!");
      }
   </script>

</head>

<body>
   <p>Click the following button to call the function</p>
   <form>
      <input type = "button" onclick = "sayHello()" value = "Say Hello">
   </form>
   <p>Use different text in write method and then try...</p>
</body>
```

Event

- JS’s interaction with HTML is handled through events that occur when the user or the browser manipulates a page.
- Event example
  - Page is loaded
  - Button is clicked
  - Pressing any keys
  - Closing window
  - Resizing

```
<form method = "POST" action = "t.cgi" onsubmit = "return validate()">
   .......
   <input type = "submit" value = "Submit" />
</form>
```

```
<input type = "button" onclick = "sayHello()" value = "Say Hello" />
```

```
<div onmouseover = "over()" onmouseout = "out()">
   <h2> This is inside the division </h2>
</div>
```

Some Standard Events

- onclick - Triggers on a mouse click
- ondrag - Triggers when an element is dragged
- ondrop - Triggers when dragged element is being dropped
- onload - Triggers when the document loads
- onkeydown - Triggers when a key is pressed
- onkeypress - Triggers when a key is pressed and released
- onmouseover - Triggers when the mouse pointer moves over an element
- onresize - Triggers when the window is resized
- onselect - Triggers when an element is selected
- onsubmit - Triggers when a form is submitted

HTML DOM

- When an HTML document is loaded into a web browser, it becomes a **document object**.
- With DOM, JS can access and change all elements of an HTML document.
- In other words: **The HTML DOM is a standard for how to get, change, add, or delete HTML elements.**

DOM Document

- If you want to access any element in an HTML page, you always start with accessing the document object.

- Finding HTML elements
  - document.getElementById(*id*)
  - document.getElementsByTagName(*name*)
  - document.getElementsByClassName(*name*)
  - document.querySelectorAll(*selectorName*);
  - document.body
  - document.images
  - document.links
  - etc.

DOM Document

- Changing HTML elements
  - *element*.innerHTML = *new html content*
  - *element.attribute = new value*
  - *element*.style.*property = new style*
  - *element*.setAttribute*(attribute, value)*

- Adding and deleting elements
  - document.createElement(*element*)
  - document.removeChild(*element*)
  - document.write(*text*)

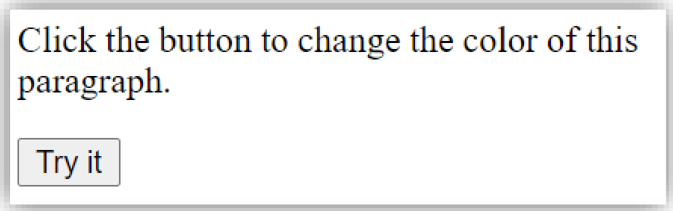
- Adding events handlers
  - document.getElementById(*id*).onclick = function(){*code*}

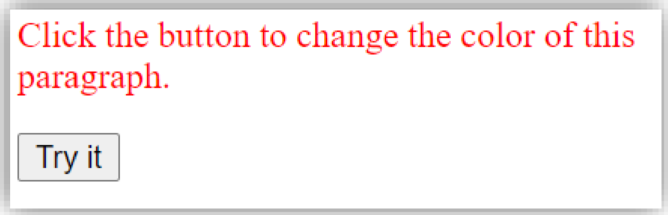
Example

- write() – writes HTML or JS code to a document.

```
document.write("<h1>Hello World!</h1><p>Have a nice day!</p>");
```

- getElementById() (by class, by classname)






```
<p id="demo">Click the button to
change the color of this paragraph.
</p>

<button onclick="myFunction()">Try
it</button>

<script>
function myFunction() {
  var x =
document.getElementById("demo");
  x.style.color = "red";
}
</script>
```

DOM Style Object

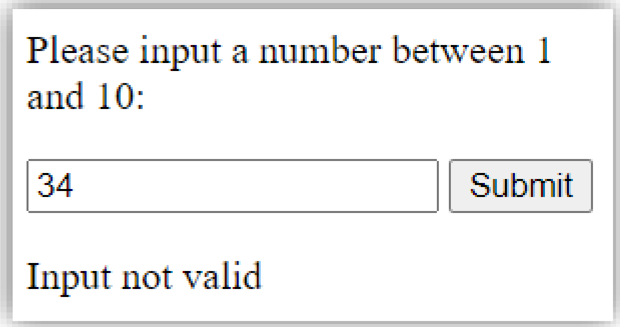
- The Style object represents an individual style statement.
- It has several properties to work such as:
  - color
  - backgroundColor
  - width
  - height
  - Etc.

- For complete list of style properties, you can refer to https://www.w3schools.com/jsref/dom_obj_style.asp

Form Validation

- JS is often used for form validation.

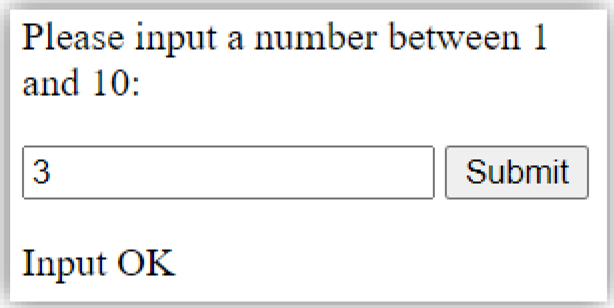
```html
<p>Please input a number between 1 and 10:</p>

<input id="numb">

<button type="button"
onclick="myFunction()">Submit</button>

<p id="demo"></p>

<script>
function myFunction() {
  var x, text;

  // Get the value of the input field with id="numb"
  x = document.getElementById("numb").value;

  // If x is Not a Number or less than one or greater
than 10
  if (isNaN(x) || x < 1 || x > 10) {
    text = "Input not valid";
  } else {
    text = "Input OK";
  }
  document.getElementById("demo").innerHTML = text;
}
</script>
```



Form Validation

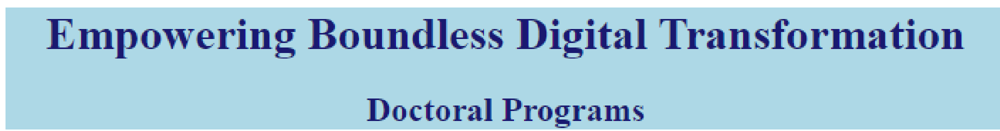
- To get the value of each input element, we can also refer to the name of form and input directly.

```
<p>Please input a number between 1 and 10:</p>
<form name="form">
<input id="numb">

<button type="button"
onclick="myFunction()">Submit</button>
</form>
<p id="demo"></p>

<script>
function myFunction() {
  var x, text;

  // Get the value of the input field with id="numb"
  x = document.form.numb.value;

  // If x is Not a Number or less than one or greater
than 10
  if (isNaN(x) || x < 1 || x > 10) {
    text = "Input not valid";
  } else {
    text = "Input OK";
  }
  document.getElementById("demo").innerHTML = text;
}
</script>
```

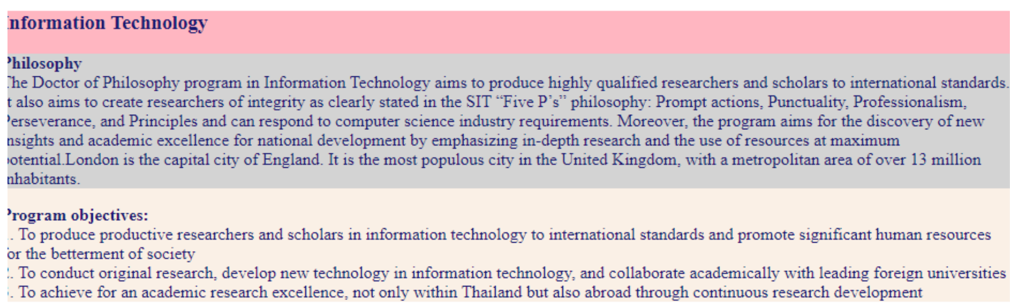


Exercise 9.1

- Using the given HTML page, modify the page according to the following requirements:
  - When a mouse is rolled over the h1, the text should be changed to “Empowering Boundless Digital Transformation”. And when a mouse is rolled out it should be changed back to original text.



  - When a mouse is rolled over the paragraph of article, the background color of paragraph should be changed to “Lightgray” and changed back to original color when a mouse is rolled out.



Exercise 9.2

Form validation exercise

- Create a form with the following data field and requirements:
  - Name – must not start with number and must not empty (Hint: you can use method charAt(0) to get the first character of String)
  - Age – must be a number and ranged from 18-150 and must not empty
  - Address – address text
  - Zip code – must be a number with 5 digits

- If one of the field is invalid, there should be an alert message appear. You can use alert function for example:
  - alert("Age input is invalid.")